MIL-A-28729B(EC)
AMENDMENT 2
26 May 1981
SUPERSEDING
AMENDMENT 1
12 April 1976

MILITARY SPECIFICATION

ANTENNA COUPLER GROUP, CU-1382( )
(HF BROADBAND, WIDE DYNAMIC RANGE)

This amendment forms a part of Military Specification MIL-A-28729B(EC) dated 15 May 1975.

PAGE 1

* 1.1, lines 1 and 2: Change "two" to "four" and line 4: Insert "and from one to sixteen" between "eight" and "radio".

* 1.2, lines 3 and 4: Delete "Type I - High linearity" and "Type II - High linearity; phase correlated" and substitute: "Type I - High linearity, 8-output
Type II - High linearity, 8-output, phase correlated
Type III - High linearity, 16-output
Type IV - High linearity, 16-output, phase correlated"

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* 3.2, line 2: Delete "Types I and II" and substitute "Types I through IV"; line 4: Insert "or sixteen" between "eight" and "receivers" and line 8: Delete "both types" and substitute "each type."

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3.2.1: Delete last sentence.

3.2.2(a): Delete "Input filter" and substitute "RF Input/bandpass filter".

* 3.2.2, after line 11: Add "The functional subdivisions shall be interchangeable as follows:

a. RF Input/bandpass filter - Types I and III; Types II and IV

b. Amplifier/Output Isolator - Types I and III; Types II and IV

c. Power Supply - All types".

3.2.6: Delete and substitute: "Non-standard parts. Approval of non-standard parts shall be in accordance with MIL-E-16400 except that prior to selection and request for approval, the procedures in MIL-STD-1631 shall be followed. Non-standard parts shall not be used in the power supply assembly with the exception of power transformers."

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3.2.15: Delete third and fourth sentences.

* 3.2.16, line 3, before "The" Insert "For Types I and II"; line 4, after "(45.72cm)" Add "and for Types III and IV, the equipment chassis shall have a maximum depth of 10 inches (25.40cm)."

* 3.2.17, line 2: Delete "It" and substitute "Types I and II"; after line 11: Add "Types III and IV shall be constructed as shown in FIGURE 1A."

* FIGURE 1, title: Insert "Types I and II" before mounting.

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* After FIGURE 1, Add the following as FIGURE 1A:

"

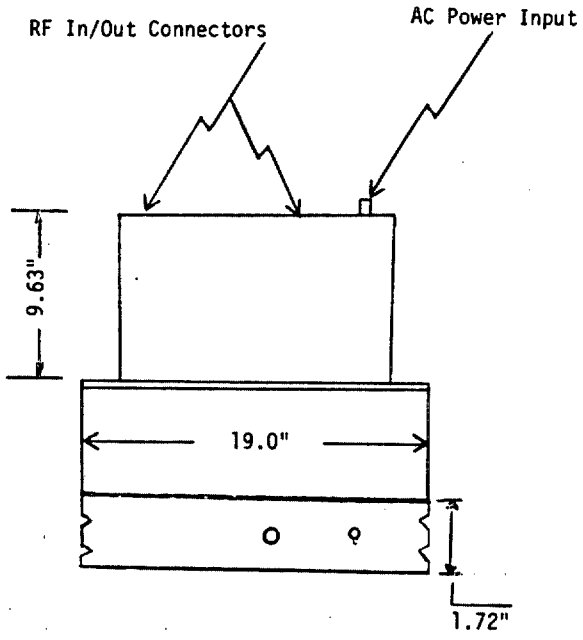


FIGURE 1A. Types III and IV mounting panel "

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3.2.29: Delete text and substitute: "Safety shall be in accordance with the Safety criteria paragraph of MIL-E-16400."

3.2.30: Delete in its entirety.

* 3.3.1: Delete text and substitute "The equipment shall be designed to achieve a minimum mean-time-between failure (MTBF) of 31,500 hours for Types I and III configurations and 34,000 hours for the Types II and IV configurations. For demonstration purposes, the specified MTBF (θ_0 as defined in MIL-STD-781) shall be 15,000 hours for each of the Types I, II, III and IV configurations."

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3.5: Delete last sentence.

* 3.5.2., line 1: Delete "The antenna coupler" and substitute "Types I and II" and after line 3, Add "Types III and IV shall provide facilities for the operation of up to sixteen radio receiving sets from a single antenna."

3.5.5, last line: Delete "." and add ", except as otherwise stated."

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* 3.5.9, line 1: change "types I and II" to "Types I through IV".

* 3.5.11, line 1: Insert "for Types I and II" between "gain" and "between"; Insert the following as the second sentence: "The insertion gain for the Types III and IV antenna couplers shall be 0 dB ±2 dB from 2 to 32 MHz. With input filter bypassed, insertion gain shall be 0 dB ±2 dB."

* FIGURE 2: Delete "Type II" and substitute "Types II and IV" and Delete "Type I" and substitute "Types I and III".

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* FIGURE 3: Delete "Type I" and substitute "Types I and III".

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* FIGURE 4: Delete "Type II" and substitute "Types II and IV".

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* 3.5.12: Insert the following as the second sentence: "The amplification of the desired signal shall not be reduced by more than 3 dB when the level of the undesired signal is increased to 2.1 volts for the Type III multicoupler and to 2.38 volts for the Type IV multicoupler."

* 3.5.16, line 1: Insert "in the Types I and II multicouplers" between "products" and "in".

* 3.5.16, line 4: Insert the following as the second sentence: "Spurious products in the Types III and IV multicouplers in the 0.3 to 32 MHz frequency range present at the outputs of the multicoupler shall be at least -65 dB relative to 0.7 volt rms when two 0.7 vrms signals are applied to the multicoupler input."

* 3.5.17: Add the following as the second sentence: "The noise figure at any frequency within the specified frequency range of the multicoupler shall be less than or equal to 11.5 dB for the Type III and less than or equal to 12 dB for the Type IV multicoupler."

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3.5.18, line 5, after "Serial Number" insert "A-17".

* 3.5.18: Insert the following as the third sentence: "Type IV multicoupler shall have phase correlation between all outputs and all other Type IV multicouplers within an absolute range of 3° over the specified frequency range of 2 - 30 MHz, input filter connected. In addition, the Type IV multicoupler shall have phase correlation with the phase standard CU-1382( ), S/N_____, within 1.5° under the same conditions."

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TABLE I and TABLE II: Add

"Surface Examination 3.2 4.4.1.16"

TABLE II, "Power output filter": Delete "output" and insert "input".

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* 4.4.1.1, line 3, after "eight" insert "and sixteen".

4.4.1.2.1: Delete in its entirety.

* 4.4.1.3, line 2: Delete "type II" and substitute "Types II and IV".

* 4.4.1.4, line 1, before "Measurements" Insert "For Types I and II,":
4.4.1.4, Delete third sentence.

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* 4.4.1.4: Add the following as the sixth sentence: "For Types III and IV, measurement shall be made with the input filter in the circuit to determine conformance with 3.5.12 with a desired signal having an input level equivalent to 140 microvolts. With the filter bypassed, the desired signal level shall be 140 microvolts. The undesired signal level initially shall be an input level equivalent to 350 microvolts."

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* 4.4.1.4.5, line 1, Delete "100 microvolts" and substitute "100 or 140 microvolts for Types I, II and Types III, IV, respectively".

4.4.1.4.8: Delete "," after "adjusted" and before "signal".

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4.4.1.6: Delete second sentence.

* 4.4.1.7, line 2, Delete "Type II" and substitute "Types II and IV".

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FIGURE 6, NOTE: Delete last sentence

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* 4.4.1.9, line 1, after "overload tests" insert "on Types I and II"; Add as the fourth sentence: "For Types III and IV, the overload voltage shall be 14 volts rms.".

4.4.1.11, first line, Insert "3.5.8 and" after "with" and insert "(CS02)" after "3.5.10".

4.4.1.11.1, second line, Insert "-10dBm" after "a"; Delete "level at any output" and substitute "at the power line input connector."

4.4.1.12.1, last line, Delete "parts" and substitute "ports".

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FIGURE 11, NOTE: Delete last sentence.

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Add new paragraph:

"4.4.1.16 Surface examination. Surface examination shall be in accordance with MIL-E-16400."

4.4.2.2: Delete last sentence and substitute "Swept frequency techniques shall be used to measure gain over the 2 to 30 MHz range.".

4.4.2.5, last line: Delete "." and add "(0.3 to 2.0 MHz measurements shall not be required)."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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